

Approach to normal & abnormal Puerperium and Lactation

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- The puerperium refers to the 6-week period following completion of the third stage of labour, when considerable adjustments occur before return to the pre-pregnant state.
- During this period of physiological change, the mother is also vulnerable to psychological disturbances, which may be aggravated by adverse social circumstances.
- Adequate understanding and support from her partner and family are crucial.
- Difficulty in coping with the newborn infant occurs more frequently with the first baby

Physiological changes

Uterine involution

- Involution is the process by which the postpartum uterus, weighing about 1 kg, returns to its prepregnancy state of less than 100 g.
- Immediately after delivery, the uterine fundus lies about 4 cm below the umbilicus or, more accurately, 12 cm above the symphysis pubis.
- However, within 2 weeks, the uterus can no longer be palpable above the symphysis.
- Involution occurs by a process of autolysis, whereby muscle cells diminish in size as a result of enzymatic digestion of cytoplasm.

- Involution appears to be accelerated by the release of oxytocin in women who are breastfeeding, as the uterus is smaller than in those who are bottle-feeding.
- A delay in involution in the absence of any other signs or symptoms, e.g. bleeding, is of no clinical significance.

Signs: causes of delayed involution

- Full bladder
- Loaded rectum
- Uterine infection
- Retained products of conception
- Fibroids
- Broad ligament haematoma

Genital tract changes

- Following delivery of the placenta, the lower segment of the uterus and the cervix appear flabby and there may be small cervical lacerations.
- In the first few days, the cervix can readily admit two fingers, but by the end of the first week it should become increasingly difficult to pass more than one finger, and certainly by the end of the second week the internal os should be closed.
- However, the external os can remain open permanently, giving a characteristic appearance to the parous cervix.

Lochia

- Lochia is the blood-stained uterine discharge that is comprised of blood and necrotic decidua.
- Only the superficial layer of decidua becomes necrotic and is sloughed off.
- The basal layer adjacent to the myometrium is involved in the regeneration of new endometrium and this regeneration is complete by the third week.
- During the first few days after delivery, the lochia is red; this gradually changes to pink as the endometrium is formed, and then ultimately becomes serous by the second week.
- Persistent red lochia offensive associated with pyrexia suggests delayed involution that is usually associated with infection or a retained piece of placental tissue.

Puerperal disorders:

Perineal complications

- Perineal discomfort is the single major problem for mothers, 80% complain of pain in the first 3 days after delivery, 25% continuing to suffer discomfort at day 10.
- Discomfort is greatest in women who sustain spontaneous tears or have an episiotomy, but especially following instrumental delivery.
- Local cooling (with crushed ice, witch hazel or tap water) and topical anaesthetics, 5% lignocaine gel, provide short-term symptomatic relief.

- Effective analgesia following perineal trauma can be achieved with regular paracetamol, diclofenac given rectally or orally may also be added.
- Codeine derivatives are not preferable, as they have a tendency to cause constipation.
- Infections of the perineum are generally uncommon.
- If there is a collection of pus, drainage should be encouraged by removal of any skin sutures.

- Spontaneous opening of repaired perineal tears and episiotomies is usually the result of secondary infection.
- Surgical repair should never be attempted in the presence of infection; (should be irrigated twice daily and healing should be allowed to occur by secondary intention).
- If there is a large, gaping wound, secondary repair should only be performed when the infection has cleared, there is no cellulitis or exudate present and healthy granulation tissue can be seen.

Obstetric palsy, or traumatic neuritis, is a condition in which one or both lower limbs may develop signs of a motor and/or sensory neuropathy following delivery.

- Presenting features include sciatic pain, foot drop, parasthesia, hypoaesthesia and muscle wasting.
- The mechanism of injury is unknown, or when herniation of lumbosacral discs (usually L4 or L5) can occur, particularly in the exaggerated lithotomy position and during instrumental delivery.
- An orthopaedic opinion should be sought and management includes bed rest with a firm board beneath the mattress, analgesia and physiotherapy.

Symphysis pubis diastasis

- Separation of the symphysis pubis can occur spontaneously in at least 1 in 800 vaginal deliveries.
- Deliberate surgical separation of the pubis in labour (symphysiotomy) is rarely performed in extreme cases of shoulder dystocia.
- Spontaneous separation is usually noticed after delivery and has been associated with forceps delivery, rapid second stage of labour or severe abduction of the thighs during delivery.
- Common signs and symptoms include symphyseal pain aggravated by weight-bearing and walking, a waddling gait, pubic tenderness and a palpable interpubic gap.
- Treatment includes bed rest, anti-inflammatory agents, physiotherapy and a pelvic corset to provide support and stability.

Thromboembolism

- The risk of thromboembolic disease rises five-fold during pregnancy and the puerperium.
- The majority of deaths occur in the puerperium and are more common after Caesarean section.
- If deep vein thrombosis or pulmonary embolism is suspected, full anticoagulant therapy should be commenced and a lower limb compression ultrasound and/or lung scan should be carried out within 24–48 hours

Puerperal pyrexia

- **Significant puerperal pyrexia is defined as a temperature of 38°C or higher on any two of the first 10 days postpartum, exclusive of the first 24 hours (measured orally by a standard technique).**
- A mildly elevated temperature is not uncommon in the first 24 hours, but any pyrexia associated with tachycardia merits investigation.
- In about 80% of women who develop a temperature in the first 24 hours following a vaginal delivery, no obvious evidence of infection can be identified.
- The reverse holds true for women delivering by Caesarean section, when a wound infection should be considered.
- Common sites associated with puerperal pyrexia include **chest, throat, breasts, urinary tract, pelvic organs, Caesarean or perineal wounds and legs**

Chest complications:

- are most likely to appear in the first 24 hours after delivery, particularly after general anaesthesia.
- Atelectasis may be associated with fever and can be prevented by early and regular chest physiotherapy.
- Aspiration pneumonia (Mendelson's syndrome) must be suspected if there is wheezing, dyspnoea, a spiking temperature and evidence of hypoxia.

Genital tract infection

- Genital tract infection following delivery is referred to as puerperal sepsis and is synonymous with older descriptions of puerperal fever, milk fever and childbed fever.

Aetiology of genital tract infections:

A mixed flora normally colonizes the vagina with low virulence.

Puerperal infection is usually polymicrobial and involves contaminants from the bowel that colonize the perineum and lower genital tract.

Organisms commonly associated with puerperal genital infection

Aerobes

- Gram-positive
 - Beta-haemolytic streptococcus, groups A, B, D
 - *Staphylococcus epidermidis* and *aureus*
 - Enterococci – *Streptococcus faecalis*
- Gram-negative
 - *Escherichia coli*
 - *Haemophilus influenzae*
 - *Klebsiella pneumoniae*
 - *Pseudomonas aeruginosa*
 - *Proteus mirabilis*

- Gram-variable

- *Gardnerella vaginalis*

Anaerobes

- *Peptococcus* sp.
- *Peptostreptococcus* sp.
- *Bacteroides* – *B. fragilis*, *B. bivius*, *B. distiens*
- *Fusobacterium* sp.

Miscellaneous

- *Chlamydia trachomatis*
- *Mycoplasma hominis*
- *Ureaplasma urealyticum*

Signs of puerperal pelvic infection

- Pyrexia and tachycardia
- Uterus – boggy, tender and larger
- Infected wounds – Caesarean/perineal
- Peritonism
- Paralytic ileus
- Indurated adnexae (parametritis)
- Bogginess in pelvis (abscess)

Symptoms of puerperal pelvic infection

- Malaise, headache, fever, rigors
- Abdominal discomfort, vomiting and diarrhoea
- Offensive lochia
- Secondary PPH

Investigations for puerperal genital infections

Investigations	Abnormalities
Full blood count	Anaemia, leukocytosis, thrombocytopenia
Urea and electrolytes	Fluid and electrolyte imbalance
High vaginal swabs	Infection screen and blood culture
Pelvic ultrasound	Retained products, pelvic abscess
Clotting screen (haemorrhage or shock)	Disseminated intravascular coagulation
Arterial blood gas	Acidosis and hypoxia (shock)

Chlamydia trachomatis puerperal parametritis may develop in one-third of women who had a preexisting infection, but presentation is usually delayed.

Common risk factors for puerperal infection

- Antenatal intrauterine infection
 - Caesarean section
 - Cervical cerclage for cervical incompetence
 - Prolonged rupture of membranes
 - Prolonged labour
 - Multiple vaginal examinations
 - Instrumental delivery
 - Manual removal of the placenta
 - Retained products of conception
 - Non-obstetric, e.g. obesity, diabetes, human immunodeficiency virus (HIV).
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- Mild to moderate infections can be treated with a broad-spectrum antibiotic, e.g. **co-amoxiclav** or a **cephalosporin, such as cefalexin, plus metronidazole**. Depending on the severity, the first few doses should be given intravenously.

Hypertension

- Nearly half of all eclamptic fits occur postnatally, and it is the highest risk time for fluid overload.
- Those with severe pre-eclampsia should be managed for the first 24 hours on a high-dependency unit until their blood pressure is controlled and they achieve a good diuresis
- Those with mild or moderate pre-eclampsia are managed on a postnatal ward until stable. They should continue any antenatally prescribed antihypertensives
- Those who are still on antihypertensives at 6 weeks postnatally should be referred for specialist assessment.

Psychiatric disorders

- Diagnosing mental illness in pregnancy is complicated by the wide variety of 'normal' emotional and behavioural changes that may occur.

Post natal physiological

- **The 'pinks':** for the first 24–48 hours following delivery, it is very common for women to experience an elevation of mood, a feeling of excitement, some overactivity and difficulty sleeping.
- **The 'blues':** as many as 80% of women may experience the 'postnatal blues' in the first 2 weeks after delivery. Fatigue, short temper, difficulty sleeping, depressed mood and tearfulness are common but usually mild, and resolve spontaneously in the majority of cases.
- **The following psychological disruptions should not be considered normal and require further assessment:**
 - panic attacks;
 - episodes of low mood of prolonged duration (>2 weeks);
 - low self-esteem;
 - guilt or hopelessness;
 - thoughts of self-harm or suicide;
 - any mood changes that disrupt normal social functioning; 'biological' symptoms (e.g. poor appetite, early wakening); change in 'affect'.

Postpartum (non-psychotic) depressive illness

- Between 10% and 15% of women will suffer with some form of depression in the first year after the delivery of their baby.
- At least 7% will satisfy the criteria for mild major depressive illness and many more could be described as having minor depression; 3–5% will suffer a severe major postnatal depressive episode.
- Without treatment, most women will recover spontaneously within 3–6 months; however, 1 in 10 will remain depressed at 1 year.
- Those with a history of depression not related to pregnancy carry between a 1:3 and 1:5 risk of a major postpartum depressive illness, while the recurrence rate of postnatal depression is as high as 50%.

Clinical features

- In contrast to puerperal psychosis, non-psychotic postpartum depression usually presents later in the postnatal period, most commonly around 6 weeks, with a more gradual onset.
- The 6-week postnatal check is an ideal opportunity to detect early postpartum non-psychotic depression
- . NICE recommends that all women are asked about their mood at least twice in the postpartum period by midwives, obstetricians, health visitors or GPs, ideally at 6 weeks and 3–4 months after the birth
- Particular attention should be paid to the assessment of women with risk factors for postnatal depressive illness.

Treatment

- Remedy of social factors.
 - Non-directive counselling.
 - Interpersonal psychotherapy.
 - Cognitive–behavioural therapy.
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- If pharmacotherapy is deemed necessary, tricyclic antidepressants or selective serotonin reuptake inhibitors (SSRIs) are appropriate.
 - There is good evidence to support the safety of the former in breastfeeding, less so for the latter. However, SSRIs in usual doses are probably safe.

Puerperal psychosis

- This very severe disorder affects between 1:500 and 1:1,000 women after delivery.
- It rarely presents before the 3rd postpartum day (most commonly the 5th), but usually does so before 4 weeks.
- The onset is characteristically abrupt, with a rapidly changing clinical picture.

Management

- The patient should be referred urgently to a psychiatrist and will usually require admission to a psychiatric unit. If possible, this should be a mother-and-baby unit under the supervision of a specialist perinatal mental healthcare team.
- These units prevent separation of the baby from its mother and this may help with bonding and the future relationship.

- Recovery usually occurs over 4–6 weeks,
- These women remain at high risk of pregnancy-related and non-pregnancy-related recurrences.
- The risk of recurrence in a future pregnancy is approximately 1 in 2, particularly if the next pregnancy occurs within 2 years of the one complicated by puerperal psychosis.
- Women with a previous history of puerperal psychosis should be considered for prophylactic lithium, started on the first postpartum day.

Postnatal visits

- postnatal care is generally now a demand-led service for low- risk women, with only 1–2 routine midwifery visits in the first 10 days and further visits as requested.
- More frequent visits will be required in high-risk women or if an abnormality has been detected, for example hypertension or maternal pyrexia.

Postnatal visits

- A formal postnatal examination is carried out at about 6 weeks postpartum
- The examination includes an assessment of the woman's mental and physical health, as well as the progress of the baby.
- In particular, direct questions should be asked about urinary, bowel and sexual function.
- Weight, urine analysis and blood pressure are checked and an examination is performed as required.
- If a cervical smear is due, it can be taken, although it is preferable to take one after 3 months postpartum.
- Contraception, pelvic floor exercises and infant immunizations are also discussed.

Contraception

- The exact mechanism of lactational amenorrhoea is poorly understood, but the most plausible hypothesis is that during lactation there is inhibition of the normal pulsatile release of luteinizing hormone from the anterior pituitary.
- Breastfeeding therefore provides a contraceptive effect, but it is not totally reliable, as up to 10% of women conceive while breastfeeding.

Contraception

- If an intrauterine contraceptive device is preferred, it is best to wait for 4–8 weeks to allow for involution.
- The combined oral contraceptive pill enhances the risk of thrombosis in the early puerperium and can have an adverse effect on the quality and constituents of breast milk.
- The progesterone-only pill (the minipill) is therefore preferable and should be commenced about day 21 following delivery.
- If done before this, there may be puerperal breakthrough bleeding. Injectable contraception, such as depot medroxyprogesterone acetate (Depo-Provera™) given 3-monthly is also very effective.
- Injectable contraception can be given within 48 hours of delivery for convenience, but it can cause breakthrough bleeding and therefore should preferably be given 5–6 weeks postpartum.

Contraception

- Sterilization can be offered to mothers who are certain that they have completed their family. Tubal ligation can be performed during caesarean section or by the open method (mini-laparotomy) in the first few postpartum days. However, it is better delayed until after 6 weeks postpartum, when it can be done by laparoscopy.
- Women who are not breastfeeding should commence the pill 3 weeks after delivery, as ovulation can occur by 4–6 weeks postpartum.

- Thank you